

10.5 Solving Problems Involving Arc Measure and Central Angles

Lesson Introduction

 Benchmark	MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.		 Learning Target	I can use the relationship between the measure of a central angle and its intercepted arc to solve problems.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	How can I use the relationship between the central angle measure and its intercepted arc to solve problems?			
 Lesson Narrative	In this lesson, students will explore central angles to discover the relationship between the arc measure and central angles. After exploration, instruction, and practice with the concept, students will then apply those concepts to solve mathematical problems involving central angles and intercepted arcs.			
 Component Summary	10.5.1 Warm-Up (~5 min.)	Noticing polygons inscribed in circles (<i>SE p. 141</i>)	10.5.4 Guided Practice (10-15 min.)	Solving problems involving central angles and intercepted arcs (<i>SE p. 145</i>)
	10.5.2 Exploration (10 min.)	Discovering the relationship between central angles and intercepted arcs (<i>SE p. 142</i>)	10.5.5 Your Turn! (5-10 min.)	Solving problems involving central angles and intercepted arcs (<i>SE p. 146</i>)
	10.5.3 Bring It Together (5 min.)	Formalizing the relationship between central angles and intercepted arcs (<i>SE p. 144</i>)	10.5.6 Wrap-Up (~5 min.)	Finding the measure of the central angle (<i>SE p. 147</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Arc - Arc Length - Arc Measure - Major Arc - Minor Arc <u>Additional Terms:</u> N/A		(Optional) Straight Edge (Optional) Markers/Colored Pencils (Optional) Compass	N/A

Teacher Tips

- Students may think the measure of an arc is half the measure of the central angle that intercepts the arc.
- Students may work 360° into their work based on what they recall about circles.
- Students may not recognize different arcs when overlap is used in a problem.

- 10.5.2 Exploration is designed to be a discovery activity for students. Be cautious not to give away the formalized information prior to the 10.5.3 Bring It Together.

Think-Pair-Share

10.5.2 Exploration provides an opportunity to leverage the Think-Pair-Share routine. In question 1, part j, students will work with a partner to share their explanations of the work they have done independently on question 1 parts a – i. This move will build confidence prior to any whole-class discussion on the relationship of central angles and intercepted arcs.

MA.K12.MTR.4.1: Discussion and Reflection

In 10.5.2 Exploration, students get to work together to understand the relationship between central angles and intercepted arcs. During this activity they will work to communicate mathematical ideas, which is a component of this MTR Standard.



Differentiate Instruction

Consider creating an anchor chart for 10.5.3 Bring It Together when students bring together their exploration and the formal introduction to the relationship between central angles and intercepted arcs. There are several different angle relationships that students learn about in this unit. Consider leveraging anchor charts to assist students with recalling the relationships based on the different components.

Lesson Notes:

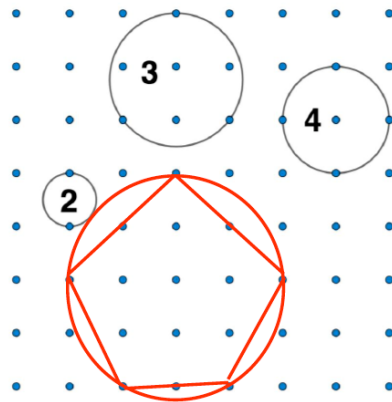
10.5.1 Warm-Up: Circles

Component Overview: Students will work with three given circles to share what they notice, and then create a polygon inscribed in a circle.



10.5.1 Warm-Up: Circles

1. A square lattice with three circles is shown where the circles pass through exactly 2, 3, or 4 points on the lattice, respectively.



- a. What do you notice about the circles?

Answers will vary. Each circle is a different size. A larger number does not correspond to a larger circle. The points evenly divide each circle into congruent arcs.

- b. Connect the points on each circle to create a polygon. Describe any pattern you notice.

Answers will vary. The line segments within each circle appear to have the same length.

- c. What type of polygon would be inside a circle that passes through 5 points?

Pentagon (5 sides)

- d. On the lattice, draw a circle that passes through exactly 5 points.

Answers will vary. On lattice.



Consider leveraging a Think-Pair-Share routine with this activity. Allow students to have some individual think time in order to try it on their own. Then, allow them to work with a partner to share their thinking and strategy. Allow student pairings to share out their answers to the entire group at the conclusion of this activity.

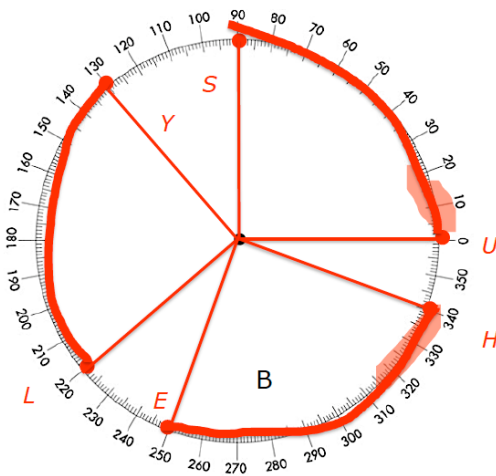
10.5.2 Exploration: Understanding Central Angles

Component Overview: Students will explore central angles using two, multi-part, scaffolded question series. For this activity, students will start working independently before sharing their answers with a partner.



10.5.2 Exploration: Understanding Central Angles

1. Circle B is shown.



- a. Find the 130° mark and label it point Y . Then, find the 220° mark and label it point L . Draw along the circumference from point Y to point L going counterclockwise.

Shown on diagram.

- b. Exactly what fraction of the circle's circumference is from point Y (130°) to point L (220°)?

$$\frac{1}{4}$$

- c. Draw the radius $\overline{BY}$ from the center B to point Y . Then, draw another radius $\overline{BL}$ from the center B to point L .

Answer on diagram.

- d. What is the measure, in degrees, of the central angle $\angle YBL$ in relation to $\widehat{YL}$?

$$90^\circ$$



Students do not need tools in order to complete this exploration. Having a straightedge and compass may be beneficial for students when they have to draw lines and arcs, although they are not constructing them. It can be helpful to keep the lines straight to avoid any visual confusion.



How did you determine the fraction of the circle's circumference in question 1 part b?

10.5.2 Exploration: Understanding Central Angles (continued)

- e. Next, along the circumference, label 0° as point U , 90° as point S , 250° as point E , and 340° as point H .

Answer on diagram.

- f. Draw two separate arcs along the circumference of the circle, one extending from point U to point S and the other extending from point E to point H . Then, draw the central angles that intersect with each part of the circumference you marked.

Answer on diagram.

- g. What is the measure, in degrees, of each of the central angles, $\angle UBS$ and $\angle EBH$?

90°

- h. Describe what you notice about all three central angles.

All three central angles are congruent.

- i. Explain the relationship between a central angle and its intercepted arc.

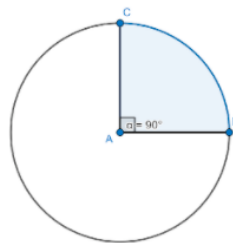
A central angle and the arc it intercepts have the same degree measure.

- j. Share your response with your partner. Listen to your partner's explanation and write a summary of the explanation. Have your partner initial stating that your summary of their explanation is accurate.

Your Partner's Explanation	Partner's Initials
<i>Answers will vary.</i>	

- k. Draw a diagram that shows the relationship between a central angle and its intercepted arc.

$m\angle CAB = 90^\circ$ and $m\widehat{CB} = 90^\circ$



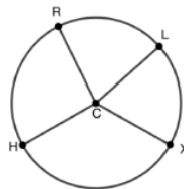
When circulating and monitoring student work, listen intently to student discussions on question 1 part j. If students are working on question 1 part k or beyond, prompt them to respond to question 1 part j in order to hear their articulation, as it is very important prior to the formal angle relationship introductions in the next activity.

10.5.2 Exploration: Understanding Central Angles (continued)

1. Discuss with your partner how arc measure relates to the relationship between a central angle and its intercepted arc. Summarize your discussion.

Answers will vary. Since a central angle represents a fraction of a circle, we know that the intercepted arc measure is the same fraction of the total circumference.

2. Circle C is shown where points R, L, X , and H are on the circle.



- a. Name the angle that intercepts $\widehat{RH}$.
 $\angle RCH$
- b. Name the arc that is intercepted by $\angle HCX$.
 $\widehat{HX}$

- c. Brooklynn and Angelina were discussing how to name an arc that is intercepted by $\angle RCX$. Each of their explanations are shown.

Brooklynn	Angelina
The arc that is intercepted by $\angle RCX$ is $\widehat{RX}$ because you just use the first and last letters of the angle to name the arc.	An arc that is intercepted by $\angle RCX$ could be $\widehat{RLX}$ or $\widehat{RHX}$ because the angle intercepts two arcs.

Discuss with your partner the conversation between Brooklynn and Angelina. Summarize your discussion.

Answers will vary. Brooklyn and Angelina agree that an arc's name must include the two points on the circle of the angle that intercepts it. Angelina notices that the measure of the angle could be either less than or greater than 180° , so there are two possible arcs that may be intercepted. Including three points in the arc's name denotes which arc is being referred to. Neither girl defined which arc is being referred to if only two points are named.



When monitoring student conversation and summaries, listen for and encourage the use of proper mathematical vocabulary.



When concluding 10.5.2 Exploration, engage students in a whole-group discussion about their response to question 2 part c. This will help highlight any misconceptions when naming arcs, which is important before moving into future lessons.

10.5.3 Bring It Together: Arc Measure and Central Angles

Component Overview: Students will be formally introduced to the relationship of a measure of an arc, and the measure of the central angle that intercepts that arc.



10.5.3 Bring It Together: Arc Measure and Central Angles

An **arc** of a circle is a section of the circumference of the circle.



The **length of an arc** is the length of the section of the circumference.



In a circle, the measure of an arc (in degrees or radians) is equal to the measure of the central angle that intercepts the arc.



In a circle, any arc whose corresponding central angle has a measure greater than 180° is considered a **major arc** and should be named using three letters.



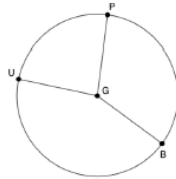
In a circle, any arc whose corresponding central angle has a measure less than 180° is considered a **minor arc** and should be named using two letters.



To provide a visual component, consider having students complete a graphic organizer to include pictorial representations of the information included in the blue boxes of this section.

10.5.3 Bring It Together: Arc Measure and Central Angles (continued)

1. Circle G has points P , B , and U on the circle with $m\angle PGB = 120^\circ$ and $m\widehat{BU} = 155^\circ$. Ryan and Katelyn are finding the measure of $\widehat{UP}$, but they found different answers.



Ryan	Katelyn
$155 + 120 = 275$	$155 + 120 = 275$
$275 - 180 = 95$	$360 - 275 = 85$
$m\widehat{UP} = 95^\circ$	$m\widehat{UP} = 85^\circ$

Do you agree with Ryan or Katelyn? Justify your answer.

Answers will vary. I agree with Katelyn because the measures of all three arcs should add to 360° in order to complete the entire circle, so Katelyn correctly subtracted the total measures of the two given arcs in order to find the third.



Question 1 is designed to demonstrate what students discovered in 10.5.2 Exploration. Use this opportunity to address student misconceptions, and ensure they have a solid understanding of the angle relationships provided.

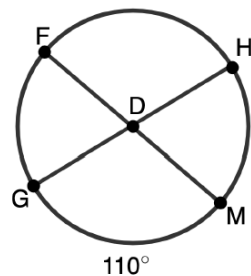
10.5.4 Guided Practice: Finding Angle Measures

Component Overview: Students will practice finding angle measures using what they have learned about central angles and intercepted angle relationships.



10.5.4 Guided Practice: Finding Angle Measures

1. Circle D has diameters $\overline{FM}$ and $\overline{GH}$ that both intersect at Point D . Given the $m\widehat{MG} = 110^\circ$, find the $m\widehat{HF}$.



110°

2. Circle T has points A , V , and C on the circle with $m\angle VTA = 21.7^\circ$. If $m\widehat{VC} = 132.3^\circ$ and $\widehat{VC}$ does not intersect $\widehat{VA}$, what is $m\angle CTA$?
3. Circle C has radii $\overline{CN}$ and $\overline{CL}$. Find the value of x if $m\angle LCN = 114^\circ$ and $m\widehat{LN} = (6x + 6)^\circ$.
4. Isabella is creating a folding fan for art class. She needs to determine the measure of the arc of the fan before she can cut the paper.

She uses the expression $(8x + 102)^\circ$ as the measure of the central angle, $\angle NFG$. For the $m\widehat{GN}$, she uses the expression $(32x + 54)^\circ$.



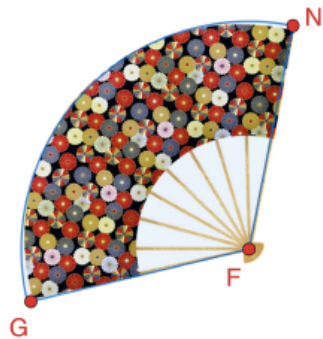
Students will be applying the relationships they learned in the prior activities to solve problems. Consider modelling question 1 for students, and sharing strategic thinking on how to approach the problem, including what information and relationships can be determined based on the information provided.



For questions 2 and 3, encourage students to draw out the circle as a visual aid when the visual is not provided in the question.

10.5.4 Guided Practice: Finding Angle Measures (continued)

What is the degree measure of the arc of the fan, $\widehat{GN}$?



$$(8x + 102)^\circ = (32x + 54)^\circ$$

$$2 = x$$

$$m\widehat{GN} = (32(2) + 54)^\circ$$

$$= 118^\circ$$



Consider finding an actual fan to enhance this activity. If available, allow students to explore this problem using a similar fan to the one provided in question 4. This can create a stronger level of engagement with this activity.

To see a visual connection to the whole circle, have students use a compass to complete the drawing of the circle created by the radius of the fan.

10.5.5 Your Turn!

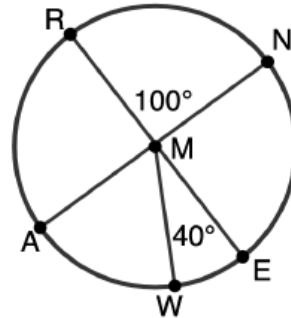
Component Overview: Students will complete two independent practice problems on solving problems using central angle relationships.



10.5.5 Your Turn!

1. In Circle M , $m\angle RMN = 100^\circ$ and $m\angle EMW = 40^\circ$. Calculate the $m\widehat{RNW}$ given that M is the center and $\overline{RE}$ and $\overline{AN}$ are diameters.

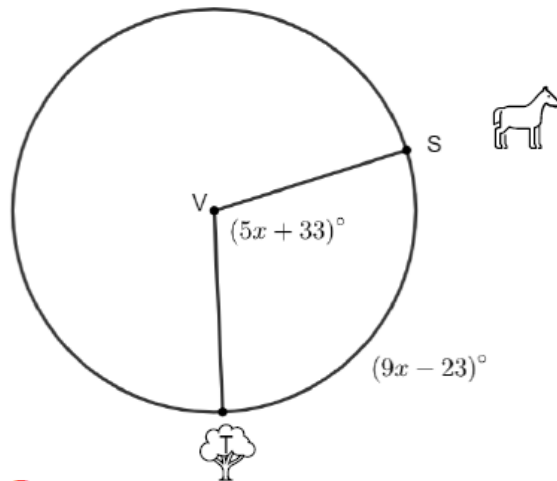
$$m\widehat{RNW} = 220^\circ$$



For students who struggled during the previous component, consider using this time to work with individual or small groups of students on angle relationships. It will be important to determine which part in the process students may be struggling with. Some students might not remember the angle relationships, and some might apply them incorrectly or make calculation errors.

2. Spirit is tied to a post at point V and is grazing in a circular pattern. He decides to walk from his position at point S to the tree at point T along the circumference of the circular area he can roam.

If the $m\angle SVT = (5x + 33)^\circ$ and $m\widehat{ST} = (9x - 23)^\circ$, find the measure of the arc that Spirit walked, $m\widehat{ST}$.



$$m\widehat{ST} = 103^\circ$$



For students who finish early, have them use the circle from question 1 to find the measure of the following arcs:

- $m\widehat{AW}$
- $m\widehat{RW}$
- $m\widehat{RNW}$
- $m\widehat{ERA}$
- $m\widehat{WNA}$

10.5.6 Wrap-Up: Ticket Out the Door

Component Overview: Students will complete a Ticket Out the Door on using the relationship between central angles and their intercepted arcs to solve problems.



10.5.6 Wrap-Up: Ticket Out the Door

- Circle X has points R , P , and A on the circle. The measure of the central angle $\angle AXP = 50^\circ$, $m\widehat{RA} = 141^\circ$, and $\widehat{RA}$ does not intersect $\widehat{AP}$. Find $m\angle RXP$.

$$m\angle RXP = m\widehat{RP} = 360^\circ - (m\widehat{AP} + m\widehat{RA})$$

$$m\angle RXP = 360^\circ - (50^\circ + 141^\circ) = 169^\circ$$



Student performance on this Wrap-Up can provide valuable feedback on how well students understood the content of this lesson.